



Agentic AI Foundation

AGNTCon + MCPCon Europe

Where the **agentic stack** is being built.

Distributed Mess: A Production Guide

To Multi-Agent Failures

Huong Vu, Databricks
September 2026

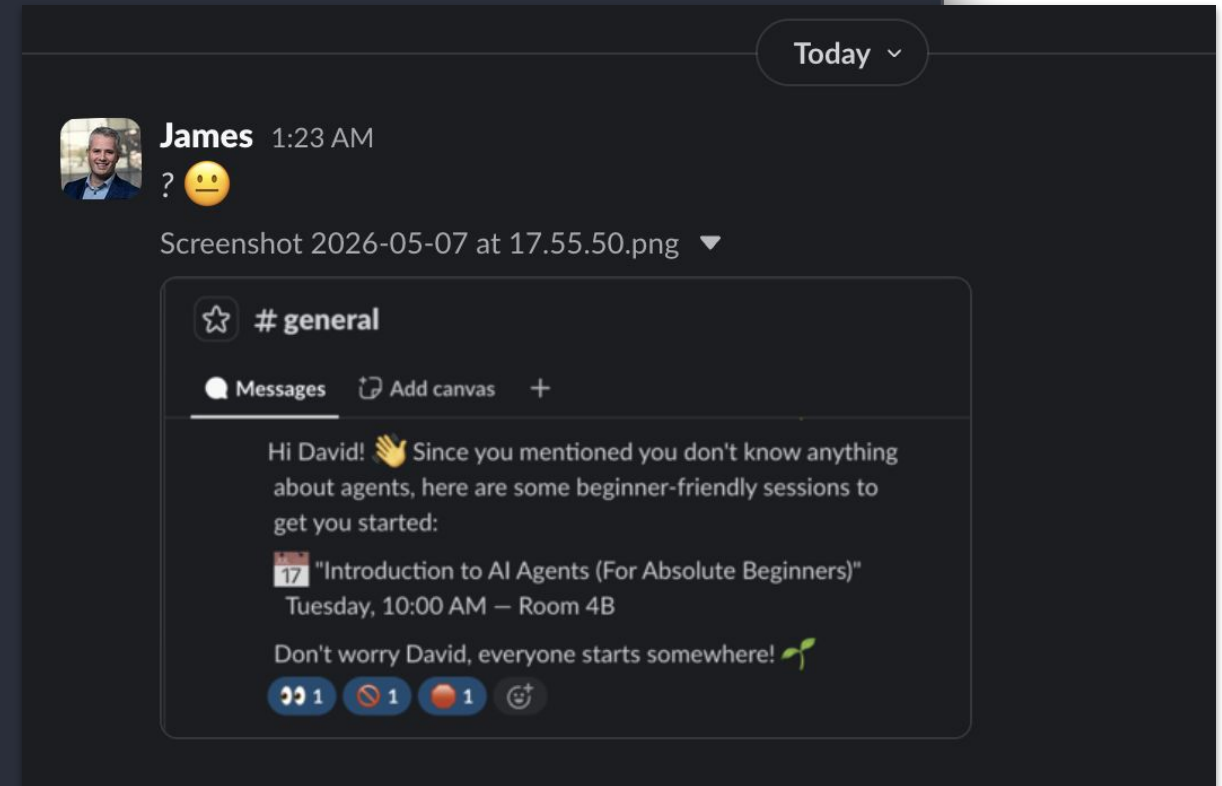
It worked in testing...

```
[→ ~ python3 ~/test_orchestrator.py
> python test_orchestrator.py

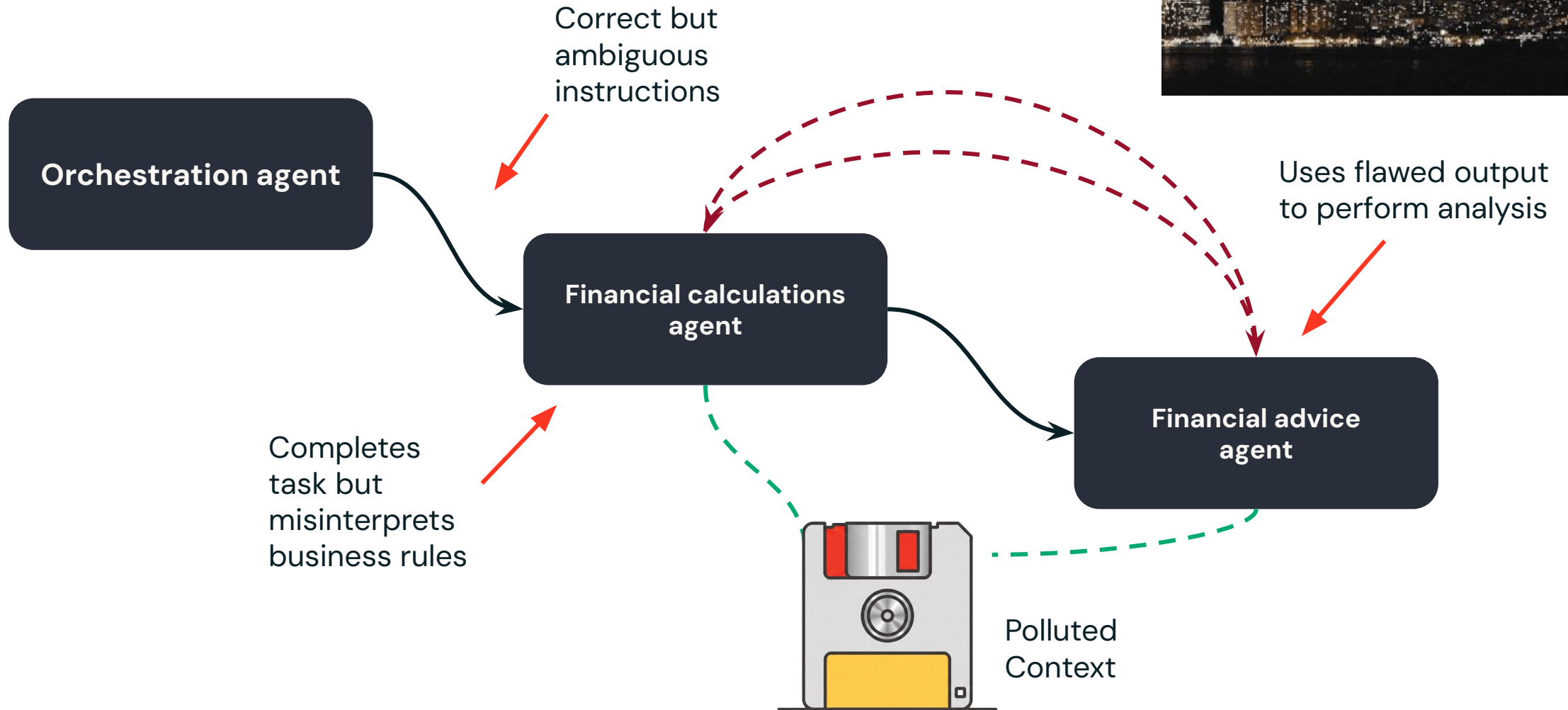
[INFO] Initializing orchestrator... ✓
[INFO] Loading BrickBot agent... ✓
[INFO] Loading tools: session_search, exhibitor_lookup,
        calendar_add, post_to_slack... ✓
[INFO] Starting test case 1/5: session recommendation

> Received query: "I don't know anything about agents,
  which sessions should I attend?" ✓
> Classifying intent... recommendation ✓
> Searching sessions by topic: "agents, beginner" ✓
> Found 4 matching sessions ✓
> Ranking by experience level... ✓
> Generating user message... ✓
> Sending confirmation to user... ✓

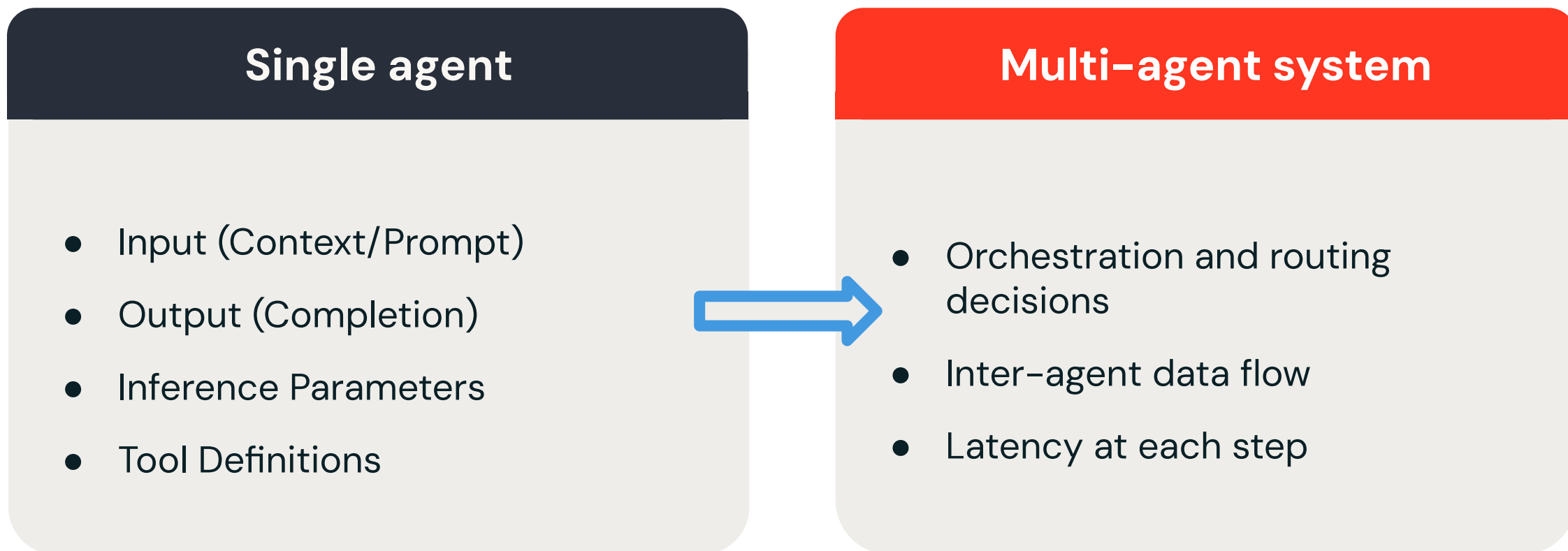
[INFO] Test case 1/5: PASSED ✓
→ ~ █
```



Multi-Agent Failure Modes



Output monitoring is no long sufficient



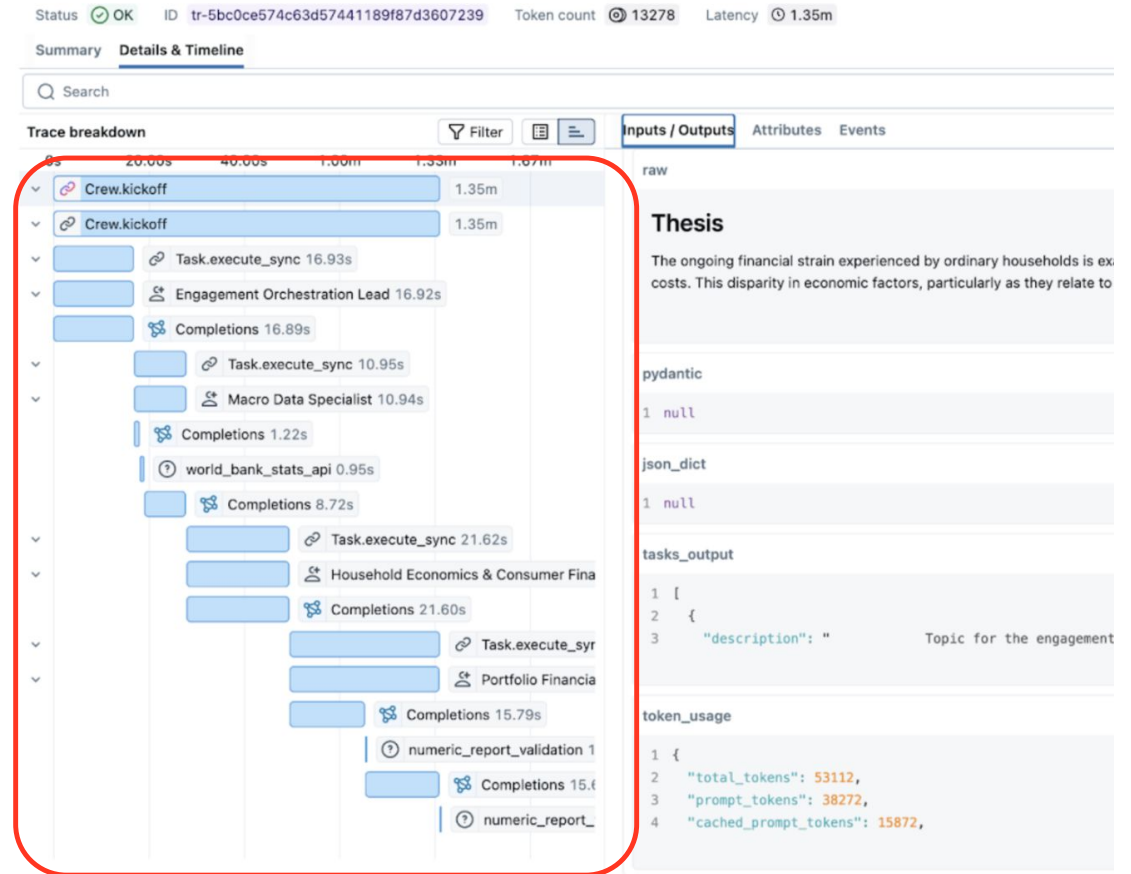
MLflow Tracing in Practice

```
import mlflow
# Turn on auto tracing by calling mlflow.crewai.autolog()
mlflow.crewai.autolog()

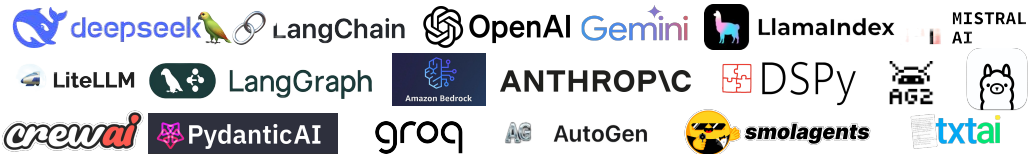
mlflow.set_experiment("/Users/huong.vu@databricks.com/crewai")

from crewai import Agent, Crew, Task
crew = Crew(
    agents = [ orchestrator, macro_data, researcher, results_lead ],
    tasks = [ plan_task, stats_task, research_task, synthesis_task ],
    verbose = True
)

result = crew.kickoff()
```



OpenTelemetry Traces



Four Dimensions of Governance

Active steering means governance shifts from dashboards to guardrails, enforced across all four dimensions.



Operational Health

Enforcing stability with circuit breakers, retries, and fallbacks across all multi-agent operations.



Cost Management

Capping spend with budgets, rate limits, and token controls across diverse LLM providers.



Compliance and Audit

Blocking violations in real time with policy gates, PII redaction, and full audit logging.



Quality and Value

Gating low-quality outputs with automated evaluation to continuously improve agent results.



AI Gateway: The First Line of Defense

The Gateway enforces Compliance and Cost and part of Operational Health

Unity Gateway

Govern and access your AI. [Learn more](#)

Models

Providers

MCPs

Skills

Agents

< > Agent Integrations

Manage usage and access for coding agents

GPT-5.6 Sol

OpenAI model, hosted on Databricks

Grok 4.6

Open source frontier model, hosted on Databricks

GLM 5.3 Flash

Open source frontier model, hosted on Databricks

Kimi K3

Open source frontier model, hosted on Databricks

Search by name or destination

Owned by me

+ Model

Name	Configuration	Total tokens (7d)	Location	Created by	Tags	Last modified
claude-opus-5		10.2M	system.ai	Databricks	—	Jul 24, 2026, 02:00 AM
gpt-5-nano		7.6K	system.ai	Databricks	—	Oct 28, 2025, 01:00 AM
claude-haiku-4-5		1.5K	system.ai	Databricks	—	Dec 19, 2025, 01:00 AM
qwen35-122b-a10b		0	system.ai	Databricks	—	Apr 23, 2026, 02:00 AM
qwen3-next-80b-a3b-instruct		0	system.ai	Databricks	—	Nov 10, 2023, 10:53 AM

Key Features

-  PII redaction
-  Rate limiting
-  Cost controls
-  Credential isolation
-  Audit logging
- ... and more

Agents → Unity AI Gateway → LLM providers



Application-Level Guardrails

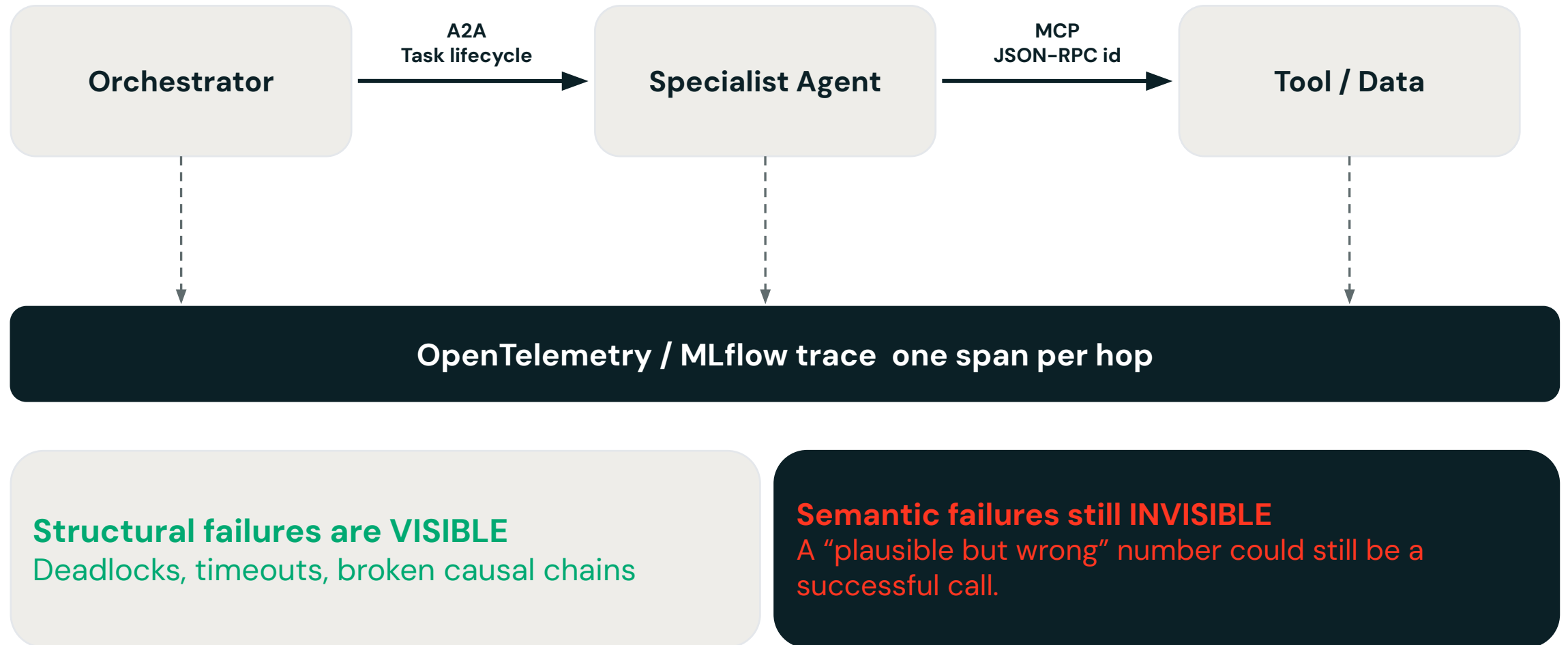
Rule: No agent can call the **Wire Transfer** tool for amounts over **\$5k** without an **MFA token**.

```
def _run(self, amount: float, account_id: str, mfa_verified: bool = False):
    span = mlflow.get_current_active_span()
    if amount > 5_000 and not mfa_verified:
        if span:
            span.set_attribute("guardrail.blocked", True)
            span.set_attribute("guardrail.reason", "MFA_REQUIRED")
            raise PermissionError("MFA required for transfers > $5,000")
    # your actual transfer logic
```



MCP/A2A Protocols as Observability Seams

Every agent boundary becomes a typed, traceable message.



Evaluations

System-Level (E2E)

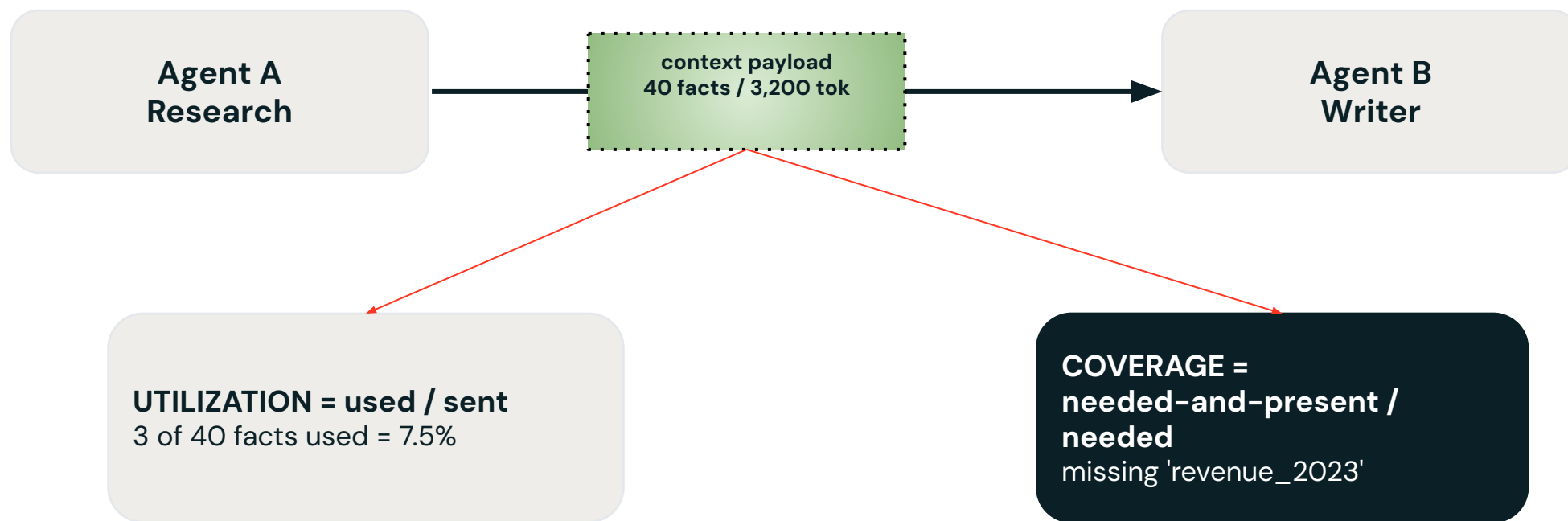
Coordination (Conflict Resolution)

Agent Handoff (Information transfer)

			Assessments (2/7)		Expectations (2/2)	
Execution t...	Request ti...	State	Relevance	Safety	expected_age...	guidelines
			PASS 76%	PASS 100%		
			Pass 76%	Pass 100%		
			Fail 24%	Fail 0%		
			null 66	null 66		
			Error 9	Error 9		
consultancy...	26.473s	04/23/2026... OK	⊖ Fail	⊙ Pass	consultancy	["Must refe
audit findin...	37.3s	04/23/2026... OK	⊙ Pass	⊙ Pass	audits	["Must cite
ny permits ...	26.879s	04/23/2026... OK	⊙ Pass	⊙ Pass	regulatory	["Must list
legal compl...	34.636s	04/23/2026... OK	⊙ Pass	⊙ Pass	legal	["Must nam
legal comp...	29.858s	04/23/2026... OK	⊙ Pass	⊙ Pass	legal	["Must con
he Chicag...	26.887s	04/23/2026... OK	⊖ Fail	⊙ Pass	inspection	["Must refe



Evaluating the Handoff: Utilization & Coverage



Sweet spot: high coverage, reasonable utilization. Either metric drifting is an early warning — catch the regression before users do.



Evaluations

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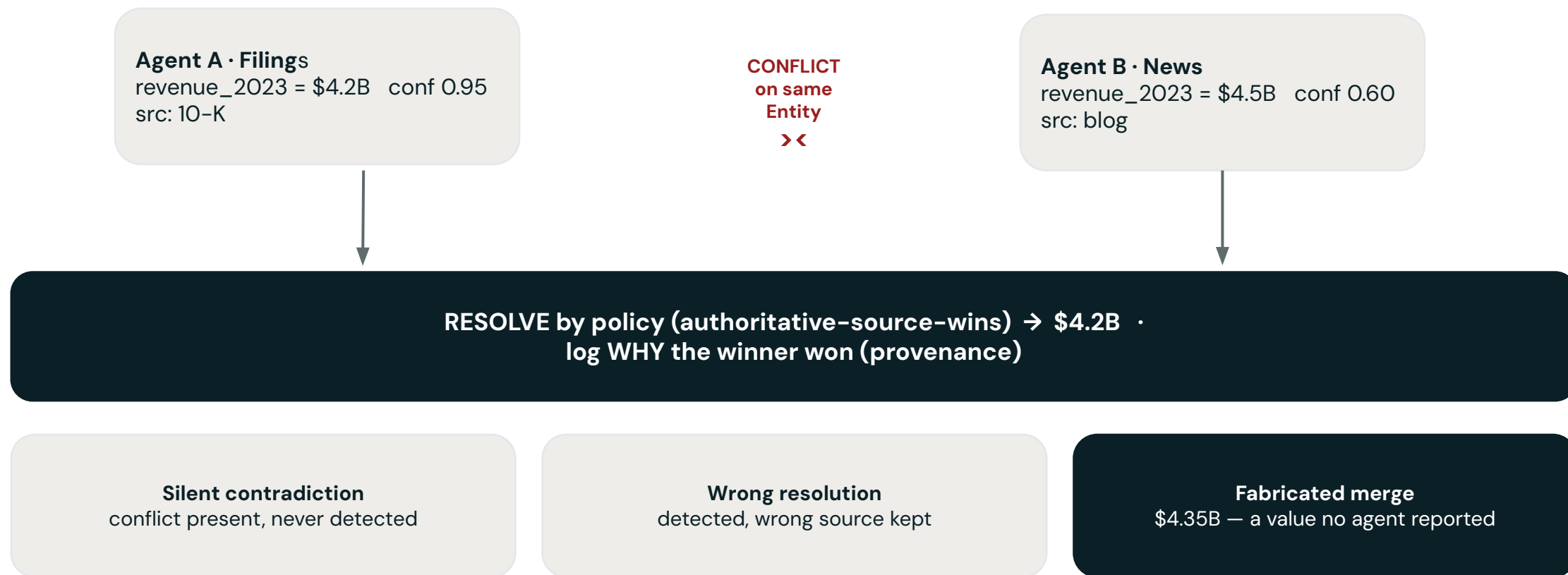
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Evaluating Coordination: The Claim Ledger

Agents emit structured claims {entity, value, confidence, source}; measure whether conflicts are detected and resolved correctly.



Evaluations

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Agent Handoff (Information transfer)

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System Level – End-to-End

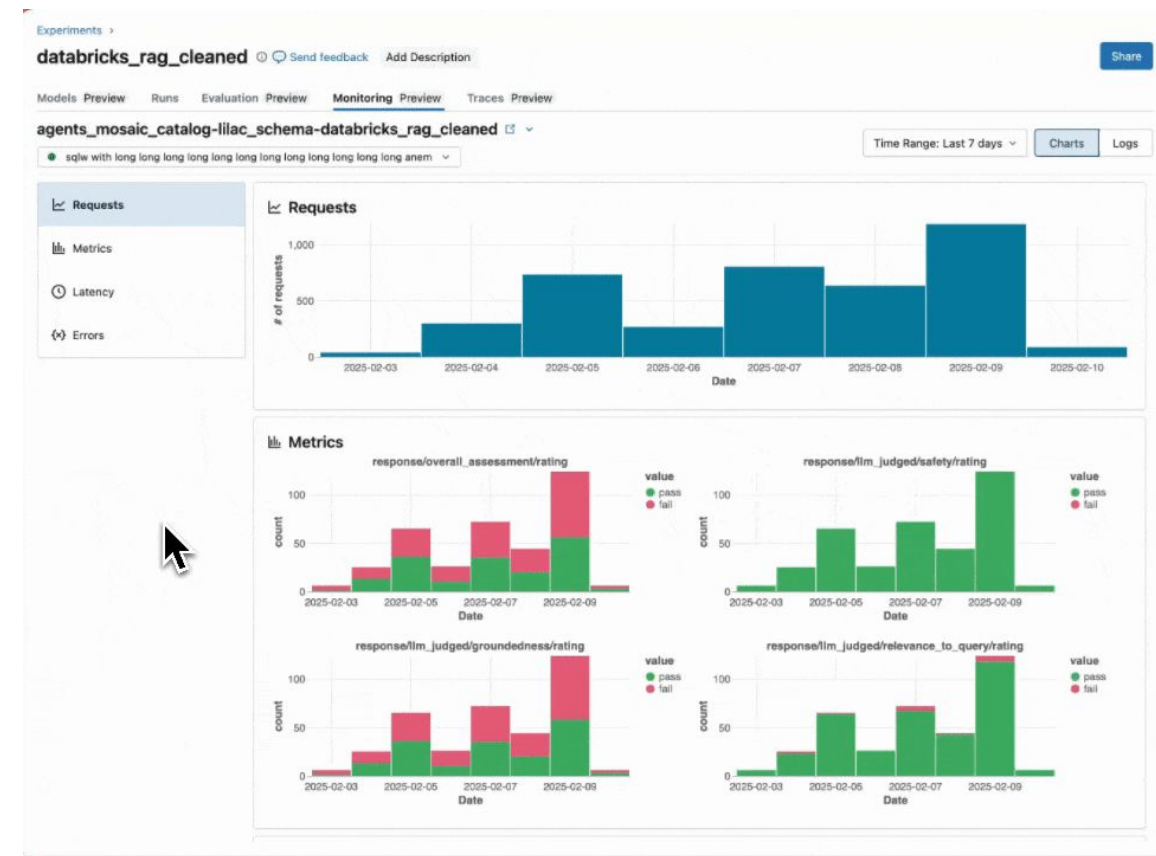
Offline evaluation

Initial quality evaluation

```
1 import mlflow
2 from mlflow.types.llm import CHAT_MODEL_INPUT_SCHEMA
3 from mlflow.models.signature import ModelSignature
4 from mlflow.models.rag_signatures import StringResponse
5
6 with mlflow.start_run(run_name="initial_agent"):
7     # Log to MLflow
8     agent_model_info = mlflow.pyfunc.log_model(
9         python_model="function_calling_agent_openai_sdk", # Agent's code
10        model_config=function_calling_agent_config.dict(), # Agent's config
11        artifact_path="agent",
12        input_example=function_calling_agent_config.input_example,
13        signature=ModelSignature(inputs=CHAT_MODEL_INPUT_SCHEMA,
14                                outputs=StringResponse),
15    ),
16
17
18
19 # Run the agent for these queries, using Agent evaluation to parallelize the calls
20 eval_results = mlflow.evaluate(
21     model=agent_model_info.model_uri, # run the logged Agent for evaluation
22     data=synthetic_eval_data, # Eval set
23     model_type="databricks-agent", # use Agent Evaluation
24 )
```

Make a change to fix retrieval quality

Online monitoring



What to Try Next

ACTION STEPS

- Instrument your multi-agent system with MLflow:
`mlflow.<flavor>.autolog()`
- Apply four dimensions of governance
- Learn more about MLflow documentation

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 /huong-vu

 /databricks-solutions/agent-control-plane

